

6(a)	<i>any five from:</i> 1 opiod binds to receptor ; 2 activates G protein ; 3 blocks Ca²⁺ channels or no / less, Ca²⁺ enter presynaptic, knob / neurone ; 4 vesicles do not, move towards / fuse with, presynaptic membrane or fewer vesicles, move towards / fuse with, presynaptic membrane ; A cell surface / plasma, membrane 5 no / less, Ach / acetylcholinesterase, released into (synaptic) cleft or no / less, exocytosis of ACh ; 6 no / less, Ach / acetylcholinesterase, binds to receptors on postsynaptic membrane ; 7 no / smaller, depolarisation of postsynaptic membrane ;	5
6(b)(i)	automatic / involuntary / no conscious thought / brain not involved ; response always the same / stereotypic ;	2
6(b)(ii)	(sensory) dorsal root ganglion or (sensory) part way along (neurone) / AW ; and (motor) CNS / spinal cord or (motor) at end (of neurone) ;	2

Question	Answer	Marks
6(b)(iii)	<i>sensory</i> 1 transmit, impulses / action potentials, from receptor ; 2 to, relay / intermediate, neurone or CNS / spinal cord or motor neurone ; <i>motor</i> 3 transmit, impulses / action potentials, from, CNS / spinal cord or relay / intermediate, neurone or sensory neurone ; 4 to effector / muscle / gland ;	4